

AKMAL FATHURRAHMAN ZULKIFLI

Operations & Product Leader | AI & Data Systems Developer

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Operations-minded product leader and AI systems developer with 3+ years at an early-stage startup, managing end-to-end platform development across hardware, software, and cloud infrastructure. Builds AI-powered internal tools and automation systems using Python, LLMs, and workflow automation. Designs scientific data pipelines for electrochemical and solid-state research including regression modelling, peak detection, and EIS fitting. Experienced in coordinating and managing remote teams across multiple time zones (UK, Afghanistan, Malaysia).

PROFESSIONAL EXPERIENCE

Platform Development Manager & Data Scientist | eProfiler Solutions Ltd | Nov 2025 – Present

Part-time | Hybrid

- Manage the end-to-end development, architecture, and feature roadmap of the ePID (eProfiler Identification & Detection) cloud platform
- Serve as the critical bridge between laboratory science and IT, translating experimental sensor data into automated cloud workflows
- Directly manage the development and integration of three core technical solutions: physical hardware systems, enterprise software platforms, and cloud infrastructure
- Manage and coordinate remote team members across multiple time zones (CSO/Research in the UK, development team in Herat, Afghanistan)
- Build and maintain SOPs to ensure consistency and efficiency across all operational workflows
- Oversee software-hardware integration and cross-functional collaboration within a small, distributed team
- Drive strategic roadmaps and project management across concurrent workstreams
- Built Python scripts for electrochemical data analysis including automated peak detection and EIS equivalent circuit fitting
- Applied and tuned regression models for parameter extraction from semiconductive-based sensor data

Production Executive & Research Data Analyst | eProfiler Solutions Ltd | May 2023 – Nov 2025

2 yrs 7 months | Part-time

- Developed and maintained end-to-end SOPs covering device assembly, quality testing, and production workflows
- Managed component procurement and inventory to support continuous device production
- Built and operated the physical device production line as a two-person team
- Conducted device testing and validation to ensure output met technical specifications
- Conducted laboratory experiments and cleaned raw experimental data for downstream analysis
- Applied regression modelling in R to extract semiconductive properties from sensor datasets
- Iteratively tuned regression models against expanded databases to improve extraction accuracy
- Presented analytical findings and model outputs to stakeholders to support project decisions

Research Assistant (Contract) | Universiti Malaya | Aug 2024 – Sep 2025

1 yr 2 months

- Managed research grant administration and compliance
- Supervised and coordinated a lab team of PhD, Master's, and undergraduate students
- Supported faculty research projects with data collection, analysis, documentation, and paper publishing

EDUCATION

PhD in Physics (Final Stage – Thesis Write-Up) | Universiti Malaya | Aug 2024 – Present

- Researching semiconductive behaviours of proteins as a method for studying denaturation which are directly applicable to biosensor development and electrochemical platform work
- Thesis write-up phase — no coursework or lab obligations

Bachelor's Degree in Physics | Universiti Malaya | 2020 – 2023

- Fundamental in mathematics, mathematical modelling, and core physics principles
- Practical grounding in digital electronics, experimental design, and instrumentation

PROJECTS

AdPulse — Python, Streamlit, pandas, Plotly, Groq API

- Multi-channel marketing analytics dashboard: auto-detects Meta / Google / SEO / social / influencer exports and produces KPIs, A–F benchmark grades, and an AI-written performance report
- Validation-first analytics with audience-fatigue detection, a CPA-weighted budget optimiser, and an LLM follow-up chat; runs fully even when the AI layer is offline (key rotation + graceful degradation)
- [Live Preview](#) and [Github Repo](#)

Teamly — Node.js, Express, Turso, Alpine.js, Resend

- Self-hosted HR portal built independently and adopted by eProfiler Solutions for internal team operations
- Leave management, WFH requests, expense claims, clock-in/out, and email-based approvals with role-based access
- [Live Preview](#) and [Github Repo](#)

PDF Triage Agent — Python, Groq API, PyMuPDF

- CLI tool that extracts structured data from a folder of PDFs into CSV + markdown report
- Schema-driven extraction — web app version in development
- [Github Repo](#)

SkillsScope — Python, Playwright, pandas, Streamlit, Plotly

- Scrapes JobStreet Malaysia for AI Engineer and Data Analyst roles and extracts in-demand skills against a 115-skill taxonomy
- Surfaces the skills each role demands, top hiring companies, and the overlap between roles in a live dashboard
- [Github Repo](#)

Further Details and Other Projects would be on [Portfolio](#).

KEY SKILLS

- Operations & Systems Design — SOP creation, process optimisation, platform architecture, production line management
- Remote Team Management — Coordinating and managing distributed teams and VAs across time zones (GMT, AFT, MYT)
- Cross-Functional Leadership — Bridging hardware, software, research, and cloud infrastructure workstreams
- Partnership & Stakeholder Management — Coordinating with labs, vendors, and external collaborators
- Communications — Stakeholder reporting, data presentations, and technical documentation
- AI-Powered Operations — Leveraging AI tools (LLMs, automation agent and coding assistant) to accelerate development, automate workflows and output scaling
- Data & Science — Regression modelling, peak detection, EIS fitting, and semiconductive property extraction; R and Python
- Startup Execution — Comfortable building from zero, wearing multiple hats, and moving fast in lean environments
- Technical Stack — Python, R, TypeScript/React, Tailwind CSS, REST APIs, Groq API, Next.js, SQLite
- AI Engineering — LLM integration, prompt engineering, structured extraction, automation agents, workflow orchestration

REFERENCES

Sara Talebi, PhD — Chief Scientific Officer (CSO), eProfiler Solutions Ltd
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